

Recovering Motion-Referring Expressions in Moving-Camera Videos via Global Motion Compensation

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ABSTRACT

Referring multi-object tracking takes a video and a natural-language expression and tracks every object the expression describes. Many expressions refer to how an object moves rather than how it looks, but common hosts read motion from raw box displacement. A moving camera corrupts that signal: a parked car sweeps across the frame and is scored as “moving,” while a car keeping pace with traffic can read as static and be dropped. Motion-language RMOT fuses dynamics with text yet leaves the camera in the signal; ego-motion compensation removes the camera but is used only for association. We connect the two with a decision-level plug-in: per-frame homographies compose into a cumulative transform, residual velocity is read at multiple gaps, a two-tower encoder aligns it with the expression, and the score is added to the host logit without host retraining. On iKUN, aligning motion with language lifts moving-class HOTA by +8.83 (native 20.25 $\rightarrow$ 29.08, $n=5$); most of that lift comes from giving the host any motion descriptor, while ego-compensation is the only engineered component with a significant effect and the only one that serves both directions of the disambiguation: removing it costs -1.77 moving ($p=0.006$) and -2.07 static-class HOTA ($p<10^{-4}$), the parked objects that raw velocity misreads as moving. Across three host-split settings, the recovered signal is monotone-inverse to native motion ability. In pooled HOTA the module matches iKUN’s published score and improves FlexHook’s V2 score outright (+0.281), while FlexHook V1 remains below its published number under a known reproduction gap.

KEYWORDS

referring multi-object tracking, ego-motion compensation, motion-language alignment, plug-in module, Refer-KITTI, video grounding

1 INTRODUCTION

Referring multi-object tracking (RMOT) outputs trajectories for the objects described by a free-form language. For example, a language description such as *left-vehicles-in-red* in the Refer-KITTI dataset is used to evaluate RMOT performance. Many of the descriptions in the Refer-KITTI dataset focus on object motion, such as “moving cars” or “a vehicle turning left”, rather than appearance. However, in videos captured by a moving camera, the observed motion is a mixture of object motion and camera motion. As a result, a stationary car may appear to move because of camera motion, while a moving car may appear stationary if it moves together with the camera.

VMRMOT [13] and its RMOT lineage [4, 10, 21] align motion with language. However, they do not explicitly distinguish object motion from camera motion. Conventional multi-object tracking (MOT) addresses this problem through camera motion compensation. For example, BoT-SORT [1] estimates camera motion, while

UCMCTrack [22] stabilizes object trajectories. However, the compensated motion is not incorporated into language-guided tracking. Camera motion can be obtained from the global positioning system (GPS) and inertial measurement unit (IMU) annotations provided by KITTI or estimated from image-based methods such as ground-plane projection. Existing RMOT methods, however, do not use this information for language-guided tracking. Instead, our method estimates camera motion directly from image features, eliminating the need for GPS or IMU sensors. This design also allows the module to be integrated into different two-stage RMOT frameworks without modifying their architectures. The only additional requirement is a single scalar calibration for each host model, as described in Section 3.4.

Global Motion Compensation-Link (GMC-Link) separates camera motion from object motion to provide motion information for language-guided tracking. It consists of four stages. First, a homography, which describes the geometric transformation between consecutive frames, is estimated to measure camera motion. The estimated homographies are accumulated over multiple frames to compensate for camera motion and provide a stable reference for measuring object motion. Second, residual velocity, which represents object motion after camera motion has been removed, is computed over multiple frames and used as a temporal motion descriptor. Third, the Contrastive Motion Aligner compares the motion descriptor with the language expression and produces a motion matching score. Finally, the motion score is combined with the host model through a calibrated decision-level correction, allowing GMC-Link to improve existing two-stage RMOT frameworks without modifying their architectures.

The accumulated homographies are the key component of GMC-Link. By compensating for camera motion, they allow the motion descriptor to represent the object’s movement rather than the camera’s movement. They also preserve motion information across multiple frames, enabling the model to capture motion patterns described by expressions such as “moving” or “turning”. Based on this design, we hypothesize that camera motion compensation is essential for distinguishing object motion from camera-induced motion. If residual velocity is replaced with raw image velocity, performance should decrease for moving objects, while stationary objects are more likely to be misclassified as moving. Section 4 evaluates this hypothesis and shows that residual velocity contributes most of the improvement for moving objects, while camera motion compensation makes the motion descriptor effective for object disambiguation rather than simply providing additional motion cues.

This work connects ego-motion compensation with motion-language modeling for referring multi-object tracking (RMOT). Our main contributions are as follows.

- We propose Global Motion Compensation-Link (GMC-Link), a plug-in module that compensates for camera

motion without modifying or retraining the host RMOT model.

- We introduce a motion representation based on residual velocity after camera motion compensation for motion-language alignment.
- We provide experimental evidence showing that camera motion compensation is the key factor for improving motion-aware RMOT.
- We demonstrate consistent improvements on iKUN and FlexHook while preserving the original detector, tracker, and appearance-language matching.

2 RELATED WORK

2.1 Referring MOT and motion-language fusion

TransRMOT [21] introduced referring multi-object tracking (RMOT) on Refer-KITTI. Later methods improved different parts of this task. For example, iKUN [4] separated language-based object matching from object tracking, while FlexHook [10] improved how visual and language information interact. ReferGPT [3] uses a multimodal large language model to describe existing object tracks, and MEX [19] provides a memory-efficient module for matching tracks with language.

More recent RMOT methods [7, 8, 11, 14] further improve how models understand object descriptions. VMRMOT [13], for example, explicitly represents motion using information such as position, direction, changes in distance, and changes in speed. However, these methods still estimate object motion from bounding-box movements in the image. When the camera moves, the observed bounding-box movement contains both the object’s own motion and the camera’s motion. Therefore, a stationary object may appear to be moving, while a moving object may appear nearly stationary. This makes it difficult for the model to correctly understand motion-related expressions.

TempRMOT [23] is different because it already stores trajectory history in recurrent memory. In our experiments, adding our external motion module to TempRMOT reduced pooled HOTA in two separate trials. This result suggests that the additional module may introduce redundant motion information or unnecessary constraints because TempRMOT already models motion over time. We therefore focus on two-stage RMOT models that do not already encode trajectory history in recurrent memory.

2.2 Camera and ego-motion compensation in tracking

Camera compensation is standard in plain MOT, where it stabilizes predictions before association. BoT-SORT [1] estimates an inter-frame transform from ORB matches [17] with RANSAC [5], while UCMCTrack [22] uses a ground-plane camera-motion model. EMAP [15] goes furthest in this direction, decoupling camera rotation and translation from object motion inside the Kalman prediction itself and improving KITTI HOTA by over 5% across four trackers. All three improve association under ego-motion, but the corrected geometry is consumed immediately by frame-to-frame

association. It is not composed into a persistent ego-stable trajectory, accumulated over the temporal span of a motion expression, or exposed as a language-aligned feature. The geometric core we rely on is classical — plane-plus-parallax motion separation [9], homography estimation [6], motion segmentation under a moving camera [2], and ego-motion estimation in visual odometry [18] — and we claim no novelty in the separation itself; the contribution is routing its output to language.

2.3 The unfilled intersection

The gap is specific: motion-language RMOT is language-aware but camera-contaminated; camera-compensated MOT is ego-stable but language-unaware. GMC-Link connects them by producing persistent ego-stable motion, aligning it with the expression, and fusing it after the host score. The method therefore preserves host appearance grounding while supplying the motion evidence that the host never had.

3 METHOD

Figure 1 shows the overall architecture of GMC-Link. The module is attached to an existing two-stage RMOT framework after object detection, tracking, and appearance-language matching have been completed. The inputs to GMC-Link are the tracked object trajectories, the corresponding video frames, and the language expression. Based on these inputs, GMC-Link estimates camera motion, extracts residual object motion, aligns it with the language expression, and produces a motion matching score. The score is then combined with the host model prediction through decision-level fusion. As illustrated in Figure 2, the complete pipeline consists of four stages: ego-motion compensation, multi-scale residual velocity extraction, motion-language alignment, and decision-level additive fusion.

3.1 Ego-motion compensation

This stage estimates camera motion from consecutive video frames so that object motion can be separated from camera-induced motion. Given the input video frames and tracked object bounding boxes, we extract and match background keypoints using Oriented FAST and Rotated BRIEF (ORB) [17]. Keypoints inside tracked object bounding boxes are removed before matching so that moving foreground objects do not affect camera-motion estimation. The remaining matched keypoints are used with Random Sample Consensus (RANSAC) [5] to estimate a homography $H_{t-1 \rightarrow t}$, which describes the geometric transformation caused by camera motion from frame I_{t-1} to frame I_t .

To describe camera motion over multiple frames, the estimated homographies are accumulated as

$$H_{t-k \rightarrow t} = H_{t-1 \rightarrow t} H_{t-2 \rightarrow t-1} \cdots H_{t-k \rightarrow t-k+1}, \quad (1)$$

where $H_{t-k \rightarrow t}$ represents the accumulated camera transformation from frame I_{t-k} to frame I_t . Instead of repeatedly transforming already warped coordinates, the accumulated homography is applied only once to the original object centroid. This reduces numerical drift and provides a stable reference for estimating ego velocity in the next stage.

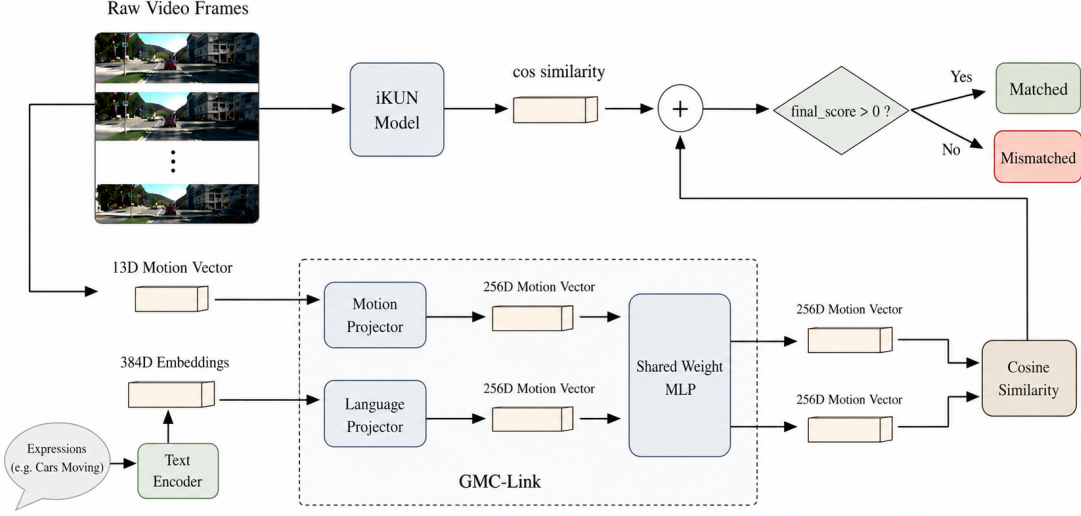


Figure 1: GMC-Link architecture. The host provides tracks and expression scores; GMC-Link estimates camera motion, derives residual motion, aligns it with language, and returns a calibrated decision-level correction.

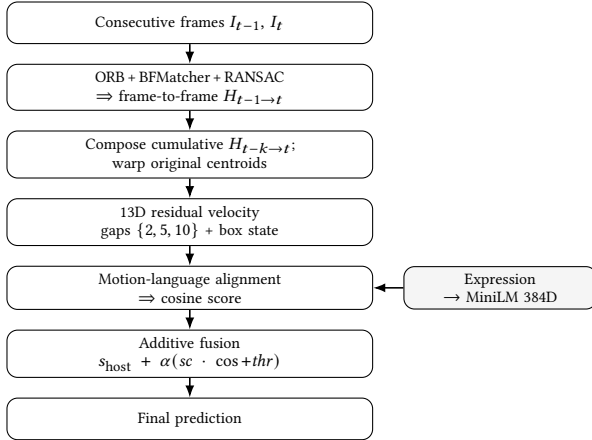


Figure 2: GMC-Link pipeline. ORB homographies compose into a cumulative transform, produce a 13D residual-velocity descriptor, align with the expression, and correct the host score through additive fusion.

3.2 Multi-scale residual velocity

Using the accumulated homography from Stage 3.1 together with the tracked object trajectories, this stage estimates residual velocity, which represents object motion after removing the estimated camera motion. For a temporal gap g , the raw velocity is the displacement of the object centroid between two frames and therefore contains both object motion and camera-induced motion. The ego velocity is obtained by applying the accumulated homography in

Eq. 1 to the previous object centroid. The residual velocity is computed as

$$v_g^{\text{res}} = v_g^{\text{raw}} - v_g^{\text{ego}}, \quad (2)$$

where v_g^{res} , v_g^{raw} , and v_g^{ego} denote the residual, raw, and ego velocities, respectively. This representation approximates the motion that would be observed by a stationary camera. For example, a parked vehicle has near-zero residual velocity even when camera motion causes a large image displacement. All velocities are normalized by the image dimensions to make them independent of image resolution.

To capture motion over different time spans, residual velocities are computed at temporal gaps of $\{2, 5, 10\}$ frames. The residual velocities are concatenated with object box information to form a 13-dimensional motion descriptor,

$$((dx_s, dy_s), (dx_m, dy_m), (dx_l, dy_l), \Delta w, \Delta h, c_x, c_y, w, h, snr), \quad (3)$$

where the first six values are the residual velocities at the three temporal gaps. The remaining features describe the object box size and location. The final feature, signal-to-noise ratio (SNR), measures the reliability of the estimated camera motion. A larger SNR indicates more reliable ego-motion compensation and provides a more stable motion representation for the following motion-language alignment stage.

3.3 Motion-language alignment

Given the motion descriptor from Stage 3.2 and the language expression, this stage measures how well the observed object motion matches the described motion. The 13-dimensional motion descriptor is encoded by a motion encoder, while the language expression is encoded by all-MiniLM-L6-v2 (MiniLM) [16] into a text embedding. Both embeddings are projected into the same feature space

Table 1: Locked fusion coefficients for Eq. 4, hand-set per host-split setting. FlexHook (V1) and (V2) are one architecture calibrated per split. Appearance scale is damped because motion is uninformative on appearance expressions.

Host	Axis	α	sc	thr
iKUN	motion	1.00	0.90	+0.17
	appearance	1.00	0.30	+0.10
FlexHook (V1)	motion	0.65	10.0	+3.0
	appearance	1.00	3.50	+0.9
FlexHook (V2)	motion	0.40	10.0	+1.3
	appearance	1.00	3.50	+1.2

using linear projection layers and a shared Multi-Layer Perceptron (MLP).

The model is trained using Information Noise Contrastive Estimation (InfoNCE) [20] loss to maximize the similarity between matched motion-expression pairs while separating unmatched pairs. During inference, the cosine similarity between the motion and language embeddings is used as the motion matching score, which is passed to the decision-level fusion stage described in Section 3.4.

3.4 Decision-level additive fusion

The motion matching score produced in Stage 3.3 is combined with the prediction score of the host RMOT model to obtain the final prediction. The final score is computed as

$$s_{\text{final}} = s_{\text{host}} + \alpha (sc \cdot \cos + thr), \quad (4)$$

where s_{host} is the prediction score produced by the host model, $\cos$ is the cosine similarity from the motion-language alignment stage, sc rescales the motion score to match the score range of the host model, thr adjusts the decision threshold, and α controls the contribution of the motion score.

The calibration parameters are selected separately for each host model because different RMOT models produce prediction scores with different ranges. The values used in this work are listed in Table 1. During inference, a keyword-based router selects either the motion branch or the appearance branch according to the language expression. Expressions containing motion-related keywords, such as *moving*, *turning*, *parked*, and *braking*, use the motion branch, while all other expressions use the appearance branch.

Since GMC-Link only modifies the final prediction score, it can be integrated into existing two-stage RMOT frameworks without changing or retraining their detector, tracker, visual encoder, or language encoder.

4 EXPERIMENTS

4.1 Setup

We evaluate GMC-Link on the Refer-KITTI V1 [21] and V2 [23] benchmarks using the HOTA metric from TrackEval [12], which jointly measures detection and association performance. Since GMC-Link is designed to improve motion grounding, we report

Table 2: Pooled HOTA on Refer-KITTI test ($n=3$, mean $\pm$ std). Δ is computed against published pooled HOTA.

Host	Native	+GMC simple	+GMC-Link	Δ paper
iKUN [4] (V1)	44.564	44.272	44.634 $\pm$ 0.066	+0.070
FlexHook (V1) [10]	53.824	53.121	53.526 $\pm$ 0.087	-0.298 [†]
FlexHook (V2) [10]	42.526	42.532	42.807 $\pm$ 0.038	+0.281

[†]FlexHook (V1) paper gap is structural (no tested configuration beats 53.824); Δ vs +GMC simple anchor = +0.405.

both moving-class HOTA, which evaluates only expressions referring to moving objects, and pooled HOTA, which evaluates all expressions in the benchmark.

To evaluate the effectiveness of GMC-Link under different host capabilities, we attach it to iKUN [4] and FlexHook [10]. We also compare against a raw-velocity variant that directly uses image-plane motion without ego-motion compensation to measure the contribution of camera motion compensation.

Results are averaged over three random seeds, while ablation studies use five seeds together with Welch’s t-test. The motion-language aligner is trained using AdamW with a learning rate of 10^{-3} , batch size 256, and 100 epochs. Camera motion is estimated using ORB with 1500 keypoints and RANSAC with a 5-pixel threshold. Both host models use their original published detection results.

Refer-KITTI is highly imbalanced, with only about one-sixth of the language expressions referring to moving objects. Consequently, improvements in motion understanding may have only a limited impact on pooled HOTA. Therefore, moving-class HOTA is used as the primary evaluation metric, while pooled HOTA is reported to reflect overall benchmark performance.

4.2 Main results

Table 2 summarizes the pooled HOTA results on Refer-KITTI. The +GMC simple baseline directly adds the motion score without calibration. Its performance is lower than the native host on iKUN and FlexHook V1, showing that simply introducing motion information is not sufficient. The motion score must be calibrated to match the score range of the host model.

After applying the proposed GMC-Link module, performance improves on all settings compared with the uncalibrated baseline. On iKUN, GMC-Link achieves 44.634 $\pm$ 0.066, which is comparable to the published result (44.564) while introducing additional motion reasoning. On FlexHook V2, GMC-Link improves pooled HOTA from 42.526 to 42.807 $\pm$ 0.038, demonstrating that the proposed module can improve overall tracking performance. Although FlexHook V1 does not exceed its published score, it consistently outperforms the uncalibrated baseline, indicating that the calibrated motion score is more effective than directly using raw motion information.

Overall, these results show that camera-motion-compensated motion information is beneficial only when it is properly aligned with the host model. More importantly, GMC-Link can improve motion reasoning while maintaining the original tracking performance of the host model, making it suitable as a plug-in module for existing RMOT frameworks.

Table 3: Effect on different host models. Native moving-class HOTA versus the moving-class HOTA gain added by GMC-Link.

Host	Native MOVING	GMC MOVING gain
iKUN [4]	20.25	$+8.83 \pm 0.83$ ($n=5$)
FlexHook (V1) [10]	43.98	$+0.91 \pm 0.79$ ($n=3$)
FlexHook (V2) [10]	48.02	$+0.43 \pm 0.17$ ($n=3$)

Moving-class denominators: 27 of 158 expressions (V1); 14.2% of 862 (V2).

Table 4: Per-class HOTA ablation on iKUN ($n=5$).

Variant	MOVING HOTA	STATIC HOTA
Native host (no module)	20.25	—
Full module	29.08 ± 0.83	43.26 ± 0.26
–ego (raw velocity)	27.31 ± 0.66	41.19 ± 0.39
–multiscale (single gap)	28.82 ± 0.84	43.42 ± 0.10
–snr	29.74 ± 0.59	43.35 ± 0.35

Native static-class HOTA under the ablation readout is not archived; deltas among module arms share one convention and are directly comparable.

4.3 Effect on Different Host Models

Table 3 compares the moving-class HOTA improvement across different host models. iKUN, which has the lowest native moving-class HOTA, achieves the largest improvement ($+8.83 \pm 0.83$). In contrast, FlexHook V1 and V2 already have stronger motion reasoning and therefore obtain smaller gains ($+0.91 \pm 0.79$ and $+0.43 \pm 0.17$, respectively).

This trend suggests that GMC-Link is most beneficial for host models with weaker motion understanding. When the host model already distinguishes camera motion from object motion reasonably well, the proposed module provides only a small improvement. Conversely, when the host struggles with motion grounding, GMC-Link can effectively compensate for this limitation and achieve a larger performance gain.

Although this trend is consistent across all evaluated settings, it is observed on two host architectures. Evaluating additional host models would help verify whether this trend generalizes to other architectures.

4.4 Ablations

Table 4 evaluates the contribution of each component in GMC-Link. The full module improves moving-class HOTA from 20.25 to 29.08 ($+8.83$). When ego-motion compensation is removed, the score decreases to 27.31, indicating that simply using image-plane motion already provides useful motion cues, but camera motion compensation is still necessary for the best performance.

Removing ego-motion compensation also reduces static-class HOTA from 43.26 to 41.19. This is because stationary objects appear to move in the image when the camera moves, causing them to be incorrectly treated as moving objects. In contrast, removing the multi-scale motion representation or the SNR weighting produces only small changes, suggesting that these components have a relatively limited impact compared with ego-motion compensation.



Figure 3: Real recovery under camera ego-motion on “moving cars” (Refer-KITTI seq 0005, frames 85/94/100 of a forward-driving clip; the green car moves, the red car is parked). The dashed line marks the mover’s image column. **Red:** the parked car’s centroid slides left as the camera passes it (center- x 261 $\rightarrow$ 104 px), so the camera-naïve host scores it “moving” (false positive), yet its ego-compensated residual is near zero and the module restores it to static. **Green:** the moving car holds its column (center- x $\approx$ 605, < 1 px/frame), so the host misses it (false negative), but its nonzero residual — unlike the static background — recovers it as moving. Both errors flip from the same host logit through the additive correction.

Figure 3 illustrates this behavior using a real example. The parked car moves across the image because of camera motion, while the moving car remains at nearly the same image location. Without ego-motion compensation, these two cases are easily confused because they have similar image-plane motion. After compensating for camera motion, the parked car has nearly zero residual motion, while the moving car retains nonzero residual motion. This allows GMC-Link to correctly distinguish between stationary and moving objects.

Finally, the complete module runs at 68 FPS (median 14.8 ms/frame, excluding host inference), indicating that the proposed module can be integrated into existing RMOT frameworks with low computational overhead.

5 CONCLUSION AND LIMITATIONS

A decision-level motion-language module is a targeted correction: it recovers the motion-class referring expressions that camera-naïve hosts miss, without touching the appearance majority. The bulk of the recovery comes from aligning a motion descriptor with language at all; ego-compensation is the component that

turns the descriptor into a disambiguator, significantly protecting both the moving and the static class. The result suggests that motion-language grounding in moving-camera video should reason over ego-stable trajectories, not raw image-plane displacement. The module keeps the tracker and detector fixed, retrains no host weights, and adds only host-specific scalar calibration. On iKUN [4] it matches published pooled HOTA while lifting the hidden motion class; on FlexHook [10] V2 it improves pooled HOTA outright.

The limits are bounded. A single ORB/RANSAC homography [1] is approximate under non-planar scenes and strong parallax; our evidence is therefore confined to planar-dominated Refer-KITTI driving footage, with no claim of transfer to handheld, aerial, or strongly non-planar video. A related failure mode is keypoint starvation: a large mover filling the frame leaves little background texture, triggering the identity fallback exactly when compensation is most needed; the snr feature down-weights such frames but does not recover them. Calibration is host-specific and benchmark-level coefficients are not universal hyperparameters. We study two host architectures across three host-split settings, with FlexHook V1/V2 not independent architectural confirmations. Hosts with native temporal memory, such as TempRMOT [23], remain out of scope on tested evidence: cascaded trials regressed pooled HOTA rather than improving it.

The broader implication is that camera compensation can serve not only tracking association, but also language grounding. Richer ego-stable trajectory descriptors are the natural next step for camera-naïve RMOT hosts.

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